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PROJECT PROPOSAL

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Memory Access Pattern Analyzer and Cache Replacement Policy Simulation

Team Members

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Project Overview:

This project focuses on building a highly configurable, trace-driven memory cache simulator. The primary goal is to evaluate how different cache configurations and replacement policies handle real-world workload traces.

To achieve this, we will first obtain memory access traces, analyze their underlying patterns (such as stride behavior and reuse distance), and then process them through our custom-built simulator.

We will compare standard replacement policies, the theoretical ideal (Belady's Optimal), and a custom replacement strategy designed by our team to see how they impact cache performance.

Key Features

- **Memory Trace Generation:**
A Pintool in C++ that captures instruction-level memory accesses (loads/stores) and logs them in a structured trace format.
- **Cache Simulator:**
A simulator supporting:
 - Cache size (capacity)
 - Block/line size
 - Set associativity
- **Belady's Optimal Policy:**
Implementation of the ideal replacement algorithm to compute the minimum possible miss rate.
- **Custom Replacement Policy:**
Design and implementation of a practical strategy for comparison against the optimal baseline.
- **Performance Analysis:**
Output metrics including:
 - Cache hit/miss rates
 - Evictions
 - Comparative performance between policies

Implementation Plan

1: Trace Extraction

- Extracting memory trace
- Develop Pin tool for memory read/write logging
- Test on programs

2: Cache Simulator & Optimal Policy

- Design cache structure (sets, tags, valid bits)
- Implement Belady's algorithm using efficient preprocessing

3: Custom Policy & Evaluation

- Implement custom replacement strategy
- Generate traces from diverse applications
- Run simulations and collect performance metrics

4: Finalization

- Analyze results and compare policies
- Optimize and document code
- Prepare final report and presentation

Expected Outcome

The project will provide insights into:

- Real-world memory access patterns
- Effectiveness of cache replacement strategies
- This project will provide a working simulator that yields concrete insights into real-world memory access patterns and the actual effectiveness of various cache replacement strategies under different hardware constraints.